Mahmoud Alyosify

NLP Engineer | Language Model Fine-Tuning, Text Understanding & Evaluation
MSc Artificial Intelligence, Queen's University (Canada)
mahmoud.alyosify@gmail.com | +20 114 555 7959 | New Cairo, Egypt · Open to Relocation & Remote
linkedin.com/in/mahmoudalyosify | github.com/MahmoudAlyosify | mahmoudalyosifysite.github.io
huggingface.co/mahmoudalyosify | kaggle.com/mahmoudalyosify

PROFESSIONAL SUMMARY

NLP Engineer working across the full language stack — **tokenisation and decoding, domain fine-tuning, information extraction, semantic search and evaluation**. Currently an **AI Engineer Intern at Injaz Tech**, building retrieval-augmented question answering with hybrid search, cross-encoder re-ranking and query rewriting. Master's research at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) studies training-free decoding-time control of LLM verbosity, evaluated on **summarisation (XSum)**, **extractive QA (SQuAD-v2)** and **open-ended generation (MT-Bench)** using **BERTScore, ROUGE, exact-match/F1** and a length-controlled LLM judge. Fine-tuned **Meta-Llama-3-8B** on a **159,826-record** domain corpus, and built an **entity-and-relation extraction pipeline** that converts unstructured open-source text into a queryable knowledge graph. Grounded in the part that decides whether NLP ships: evaluation that does not flatter the model.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

NLP Core	Transformers, Tokenisation, Text Summarisation, Extractive Question Answering, Information & Relation Extraction, Knowledge-Graph Construction, Text Classification, Semantic Similarity & Redundancy Detection, Sentence Embeddings, Semantic Search, RAG, Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Re-Ranking, Query Rewriting
Language Models	Instruction Fine-Tuning, LoRA, QLoRA, PEFT, Custom Decoding Strategies, Prompt Engineering, Structured Outputs, Self-Hosted Inference · *Mistral-7B, Llama-3.1-8B, Meta-Llama-3-8B, Qwen2.5-3B*
Frameworks	PyTorch, Hugging Face Transformers / Datasets / PEFT, TensorFlow, Scikit-learn, NLTK, LangChain, LangGraph, sentence-transformers, FAISS, ChromaDB, Pandas, NumPy
Evaluation	BERTScore, ROUGE, Exact-Match / F1, LLM-as-Judge, MT-Bench, XSum, SQuAD-v2, Non-Inferiority Testing, Ablation Studies, Reproducible Benchmarking
Deployment	AWS, Docker, FastAPI, Streamlit, ONNX, CI/CD, Git/GitHub, Linux, SQL
Programming	Python, SQL, C++, C#, JavaScript, R, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built retrieval-augmented question answering over internal documentation with **LangChain** and **LangGraph**: header-aware Markdown chunking, sentence embeddings in **FAISS**, BM25 and hybrid RRF search, **cross-encoder re-ranking**, query rewriting and follow-up question contextualisation for multi-turn chat.
- Benchmarked retrieval with custom metrics (top-1 accuracy, MRR), raising top-1 accuracy from **75% to 94%**, and built an evaluation suite (LLM-as-judge correctness and groundedness plus deterministic key-fact checks) that runs as a CI regression gate.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Cairo, Arish & Assiut, Egypt

- Teach **Machine Learning** (90-hour bootcamps in Cairo, Arish and Assiut) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED NLP PROJECTS

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation *2026 – Present*
MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi

- Research on **training-free decoding-time control** of LLM verbosity, tested against plain truncation on **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B** (MT-Bench, XSum, SQuAD-v2). An entropy-gated **logits processor** cut **25.5–34.9%** of MT-Bench tokens within pre-registered BERTScore margins, but an audit showed the saving was truncation.
- A survey of **23** training-free mechanism families found none that beat truncation without oracle labels. The proposed **Entailment-Licensed In-Decoding Elision** (sentence encoder + NLI model) beats truncation on every model, removing 0.9–2.2% of tokens; scoping it to discourse units cut reader-judged harmful elisions from 49–69% to **10.2%**.

Horus-OSINT — Domain-Adapted Language Model for Event Intelligence *2026*
Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python

- Curated and formatted **159,826** GTD/GDELT records into an instruction dataset, then fine-tuned **Meta-Llama-3-8B** for natural-language querying of global event data.
- Deployed the adapted model on **AWS** as an interactive conversational assistant over structured and unstructured intelligence sources.

HORUS Sentinel — Text-to-Knowledge-Graph Intelligence Pipeline *2026*
Information Extraction · Knowledge Graphs · Multi-Agent Systems · Fine-Tuned LLM

- Built a pipeline that turns continuously collected **unstructured open-source text** into a **living knowledge graph**, correlating entities and relations across sources rather than treating each document in isolation.
- A self-hosted fine-tuned model reasons over the graph to generate **prioritised, evidence-backed reports** with every claim traceable to source text.

Automated Multimodal Agent — Document Understanding to Narrated Output *2026*
Document Parsing · Text Generation · Speech Synthesis

- Built an autonomous pipeline that extracts and restructures document content into slide-level summaries, then renders them as narrated audio — chaining extraction, generation and synthesis end to end.

Additional Projects

- **SimCLR-Vision-SSL** — contrastive representation learning with **FAISS** embedding retrieval — the same vector-search machinery used in semantic search and RAG.
- **Human vs. Agentic Pull Requests** — large-scale text and metadata mining of GitHub pull-request discussions.
- **Also:** AudioShield (audio classification) · SCRAP · RL-Cloud-Autoscaler · Vitalism Solution (rPPG).

CERTIFICATIONS & AWARDS

AWS Academy — Machine Learning for NLP
Generative AI & Prompt Engineering — IBM / Coursera
HCIP – AI (Huawei)
1st Place — Smart Cities Hackathon 2022
AWS Academy — ML Foundations, Data Engineering
Machine Learning Specialization — DeepLearning.AI
Digilians Presidential Scholarship — Queen's University
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.